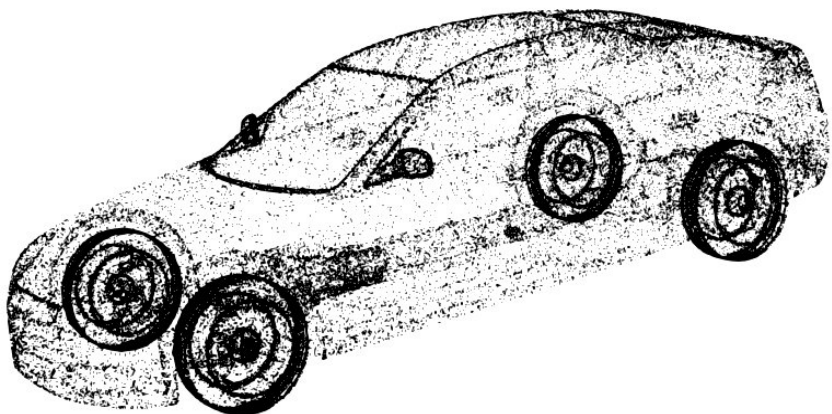


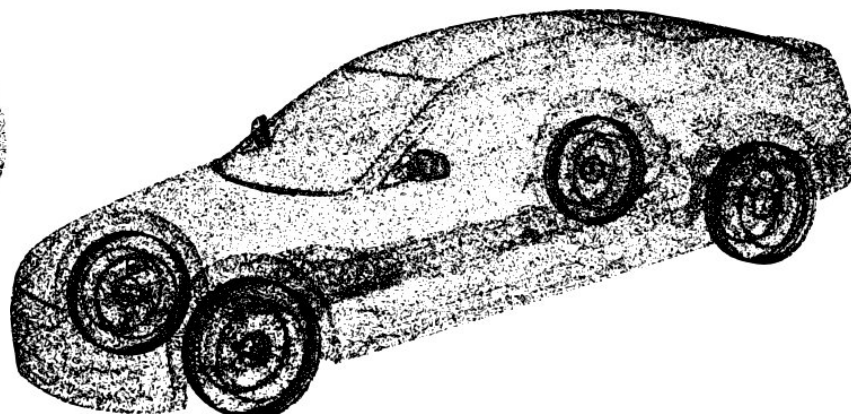
# Meeting 25.06.2026

Update with Parsa

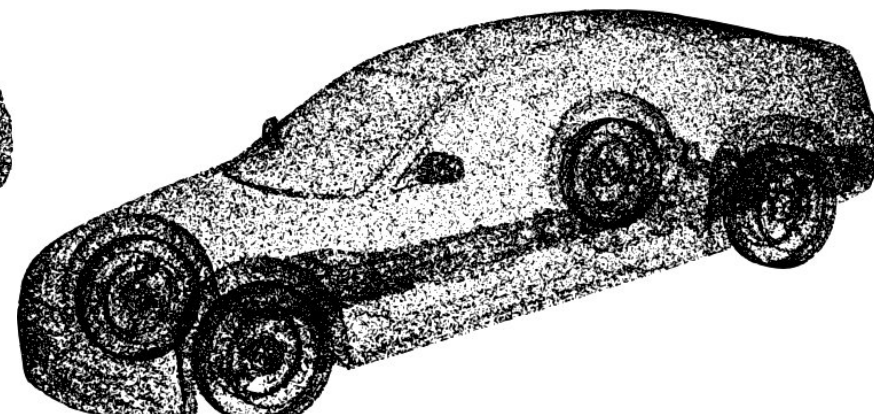
# The density-based point sampling



random distribution  $\beta=0$



Density correction intensity  $\beta=0.5$



Density correction intensity  $\beta=1.0$

The algorithm first estimates a **log-density** for each point based on its local k-nearest neighbors (typically  $k=24$ ):

$$\log \rho_j = -\log(d_k^{(j)})$$

Once the log-densities are calculated for all points, the sampling weights and final probabilities are derived:

$$w_i = \rho_i^{-\beta}$$

$$p_i = \frac{w_i}{\sum_j w_j}$$

# Training - Loss Formulation

- The total loss balances standard supervised learning across sampling strategy of random ( $\beta=0$ ) and density inverse ( $\beta=1$ ) with a consistency penalty. This forces the model's predictions to remain invariant despite the spatial density shift.

$$L_{\text{primary}} = L(\widehat{y}_{\beta=0}, y_{\text{gt}})$$

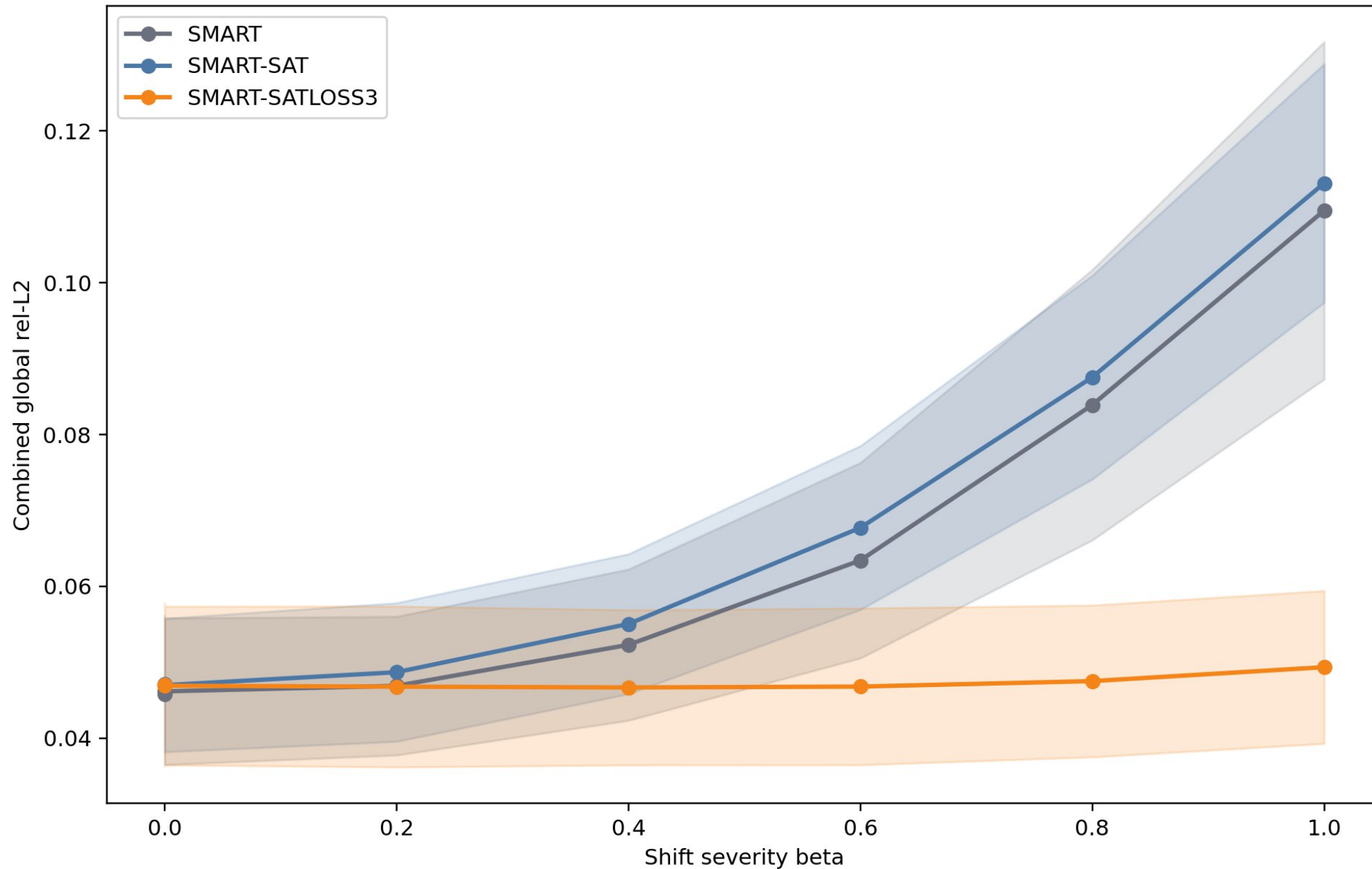
$$L_{\text{secondary}} = L(\widehat{y}_{\beta=1}, y_{\text{gt}})$$

$$L_{\text{pred-match}} = L(\widehat{y}_{\beta=0}, \widehat{y}_{\beta=1})$$

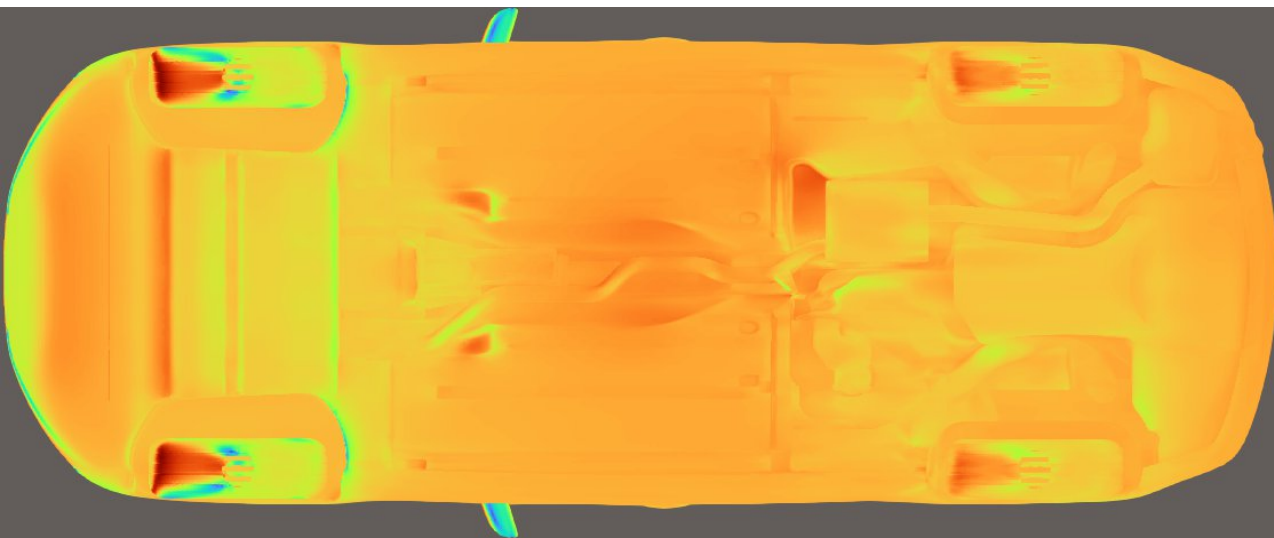
$$L_{\text{total}} = \text{Sum}(L_{\text{primary}}, L_{\text{secondary}}, L_{\text{pred-match}})$$

Future: Trying ConFIG and GradNorm

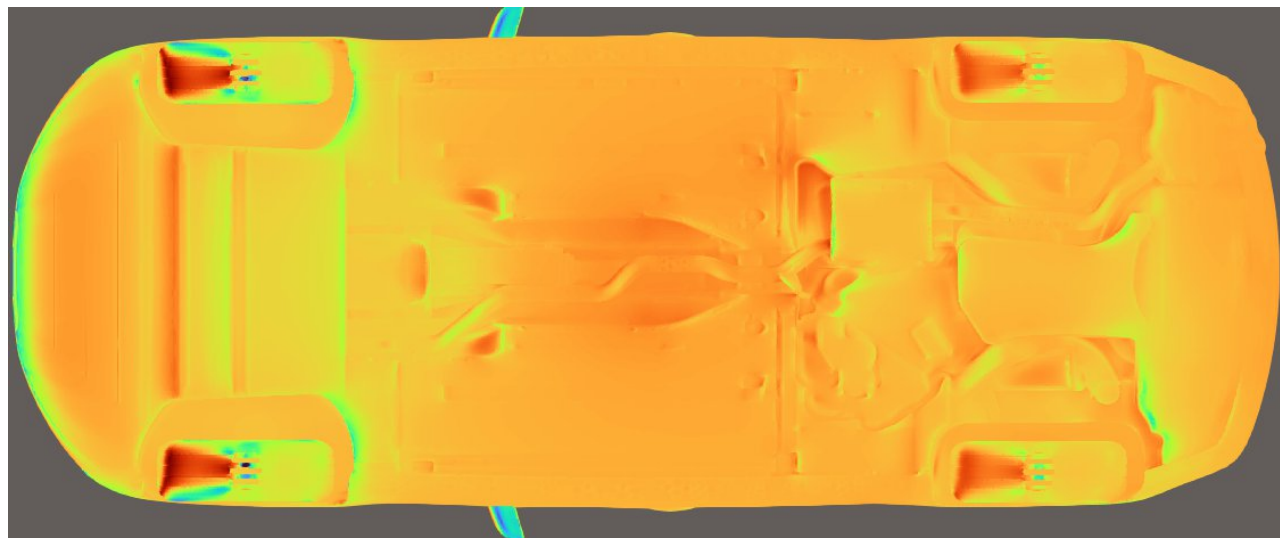
## GMP Results – test across 20 test samples with total of 6 Betas (4 unseen)



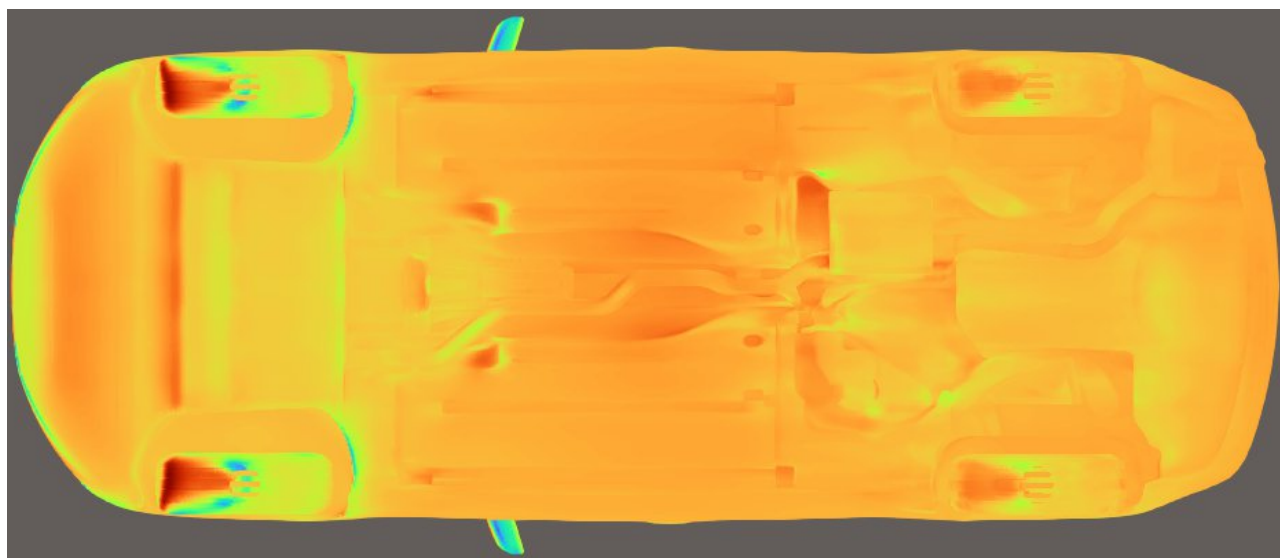
SMART\_SAT



GT



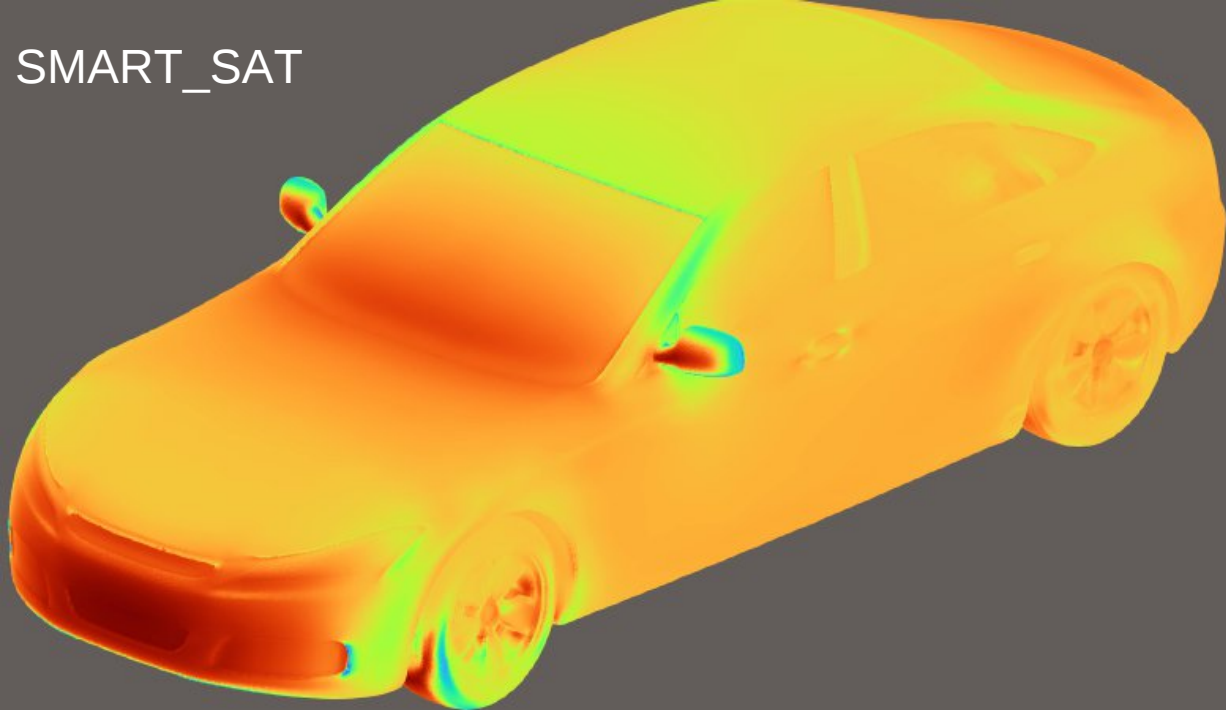
SMART\_SATLOSS



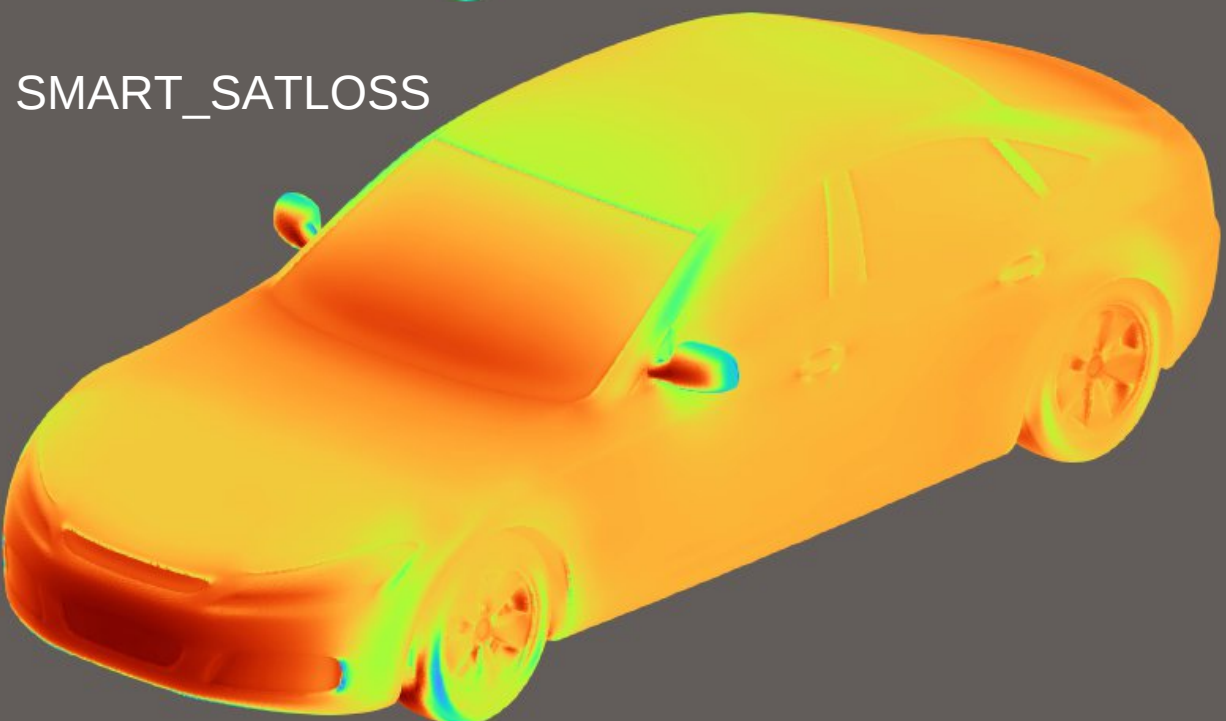
SMART



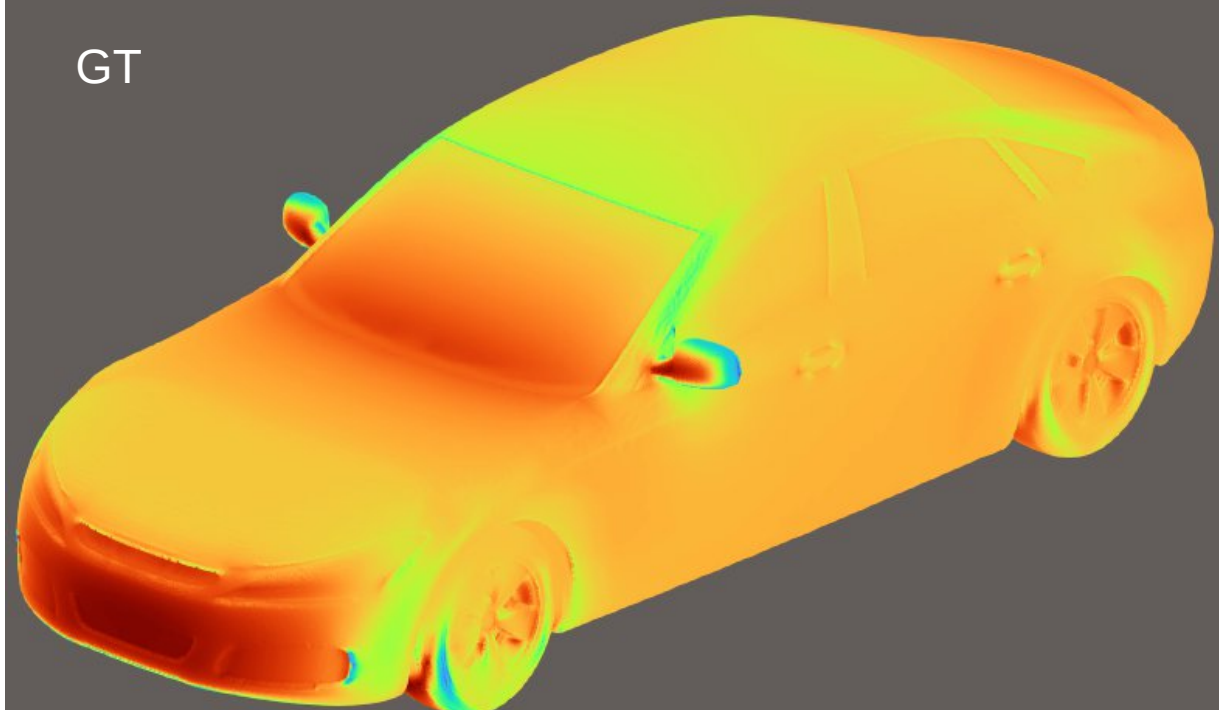
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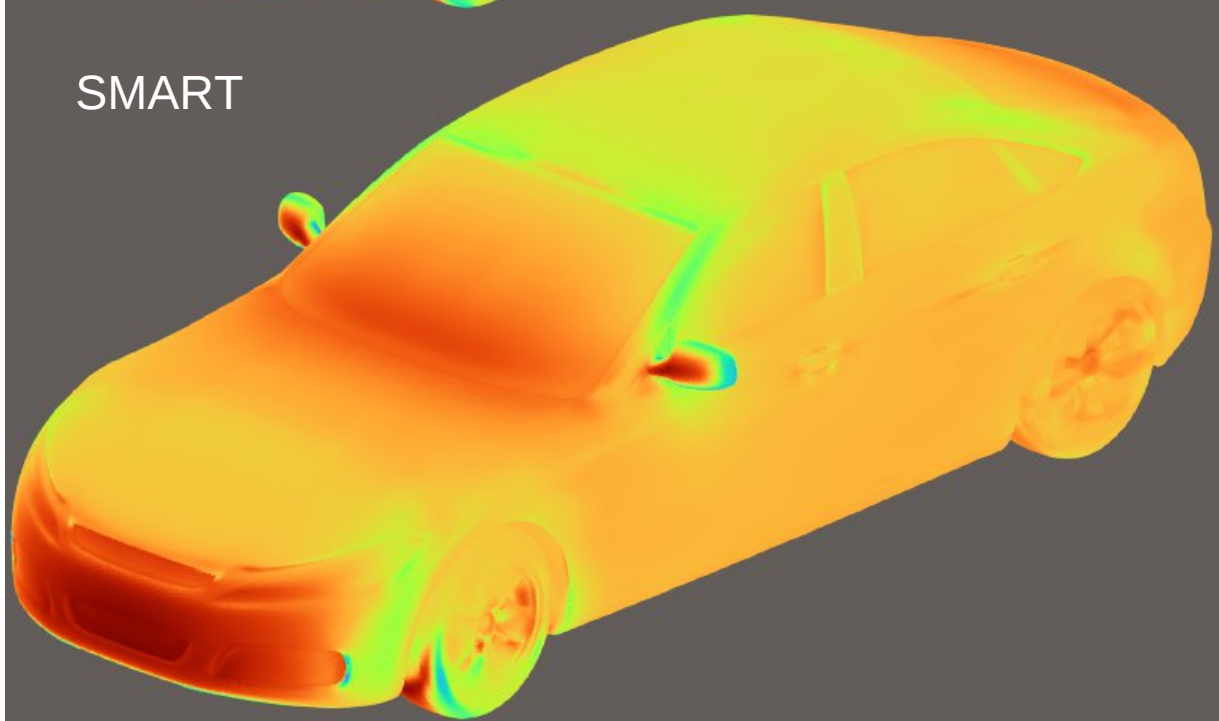
SMART\_SATLOSS



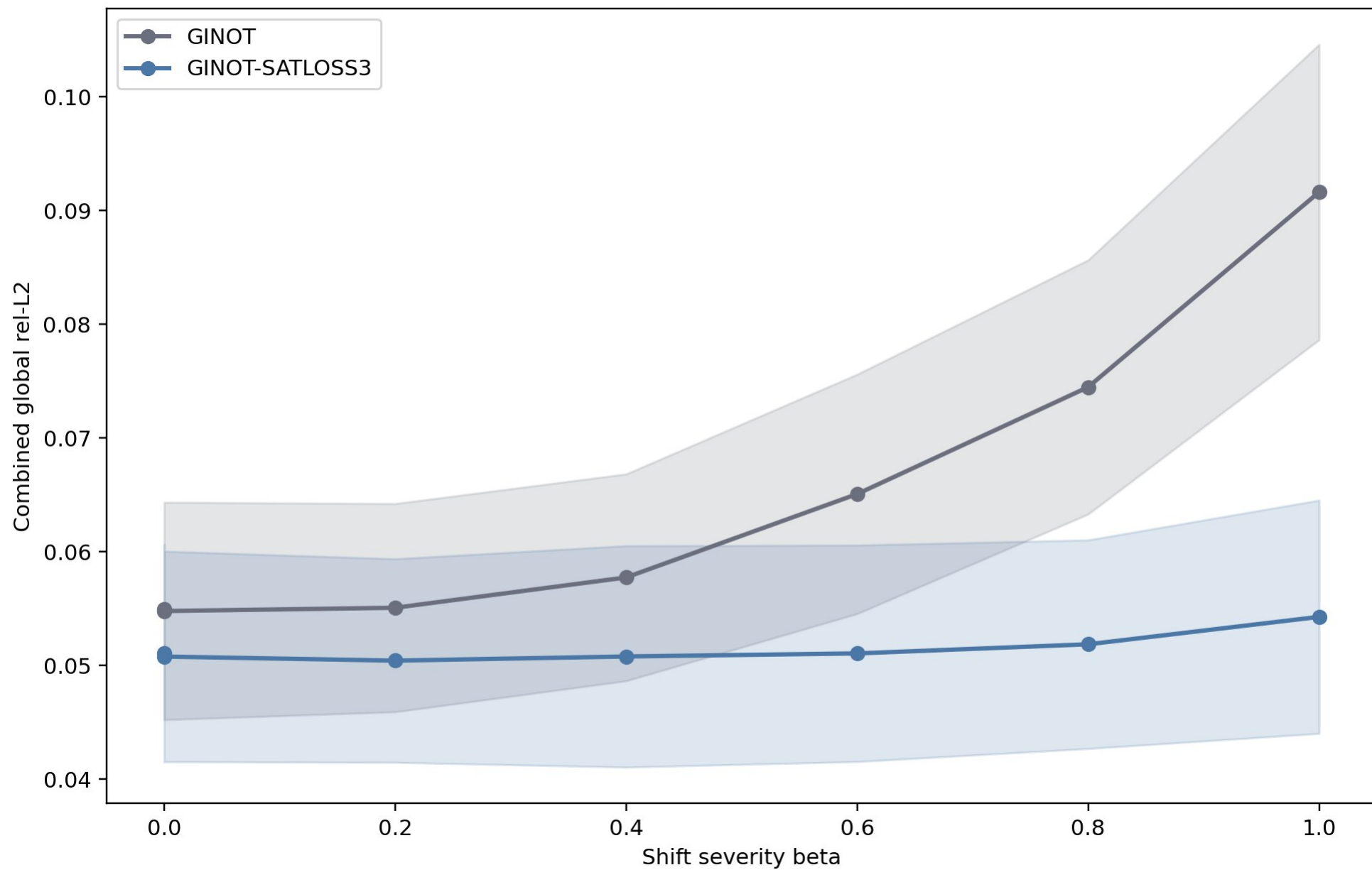
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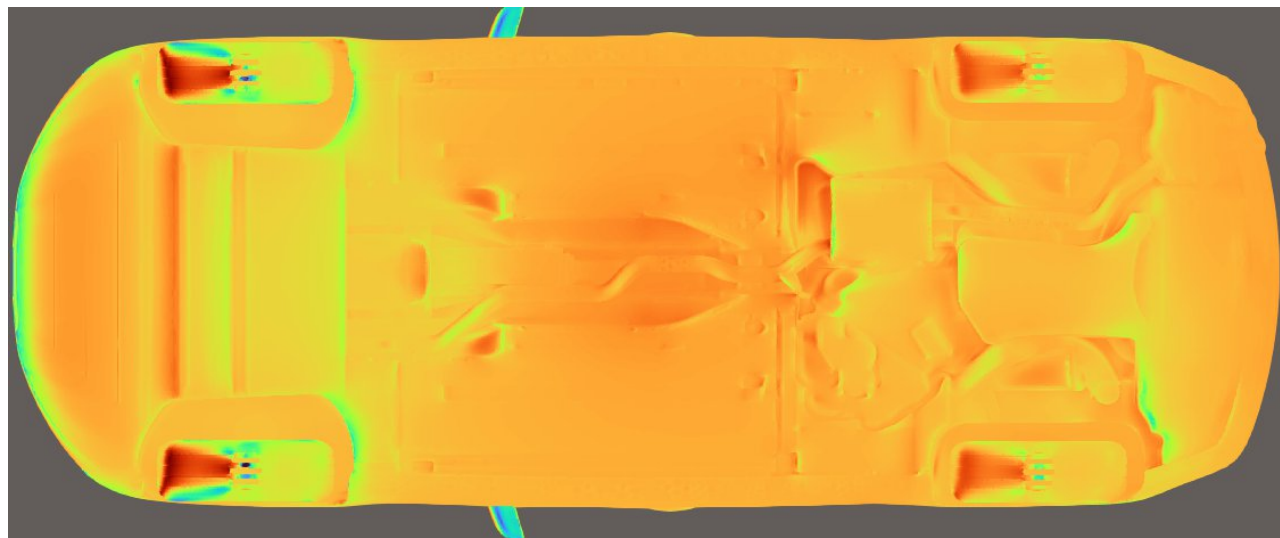
# GINOT Results – test across 20 test samples with total of 6 Betas (4 unseen)



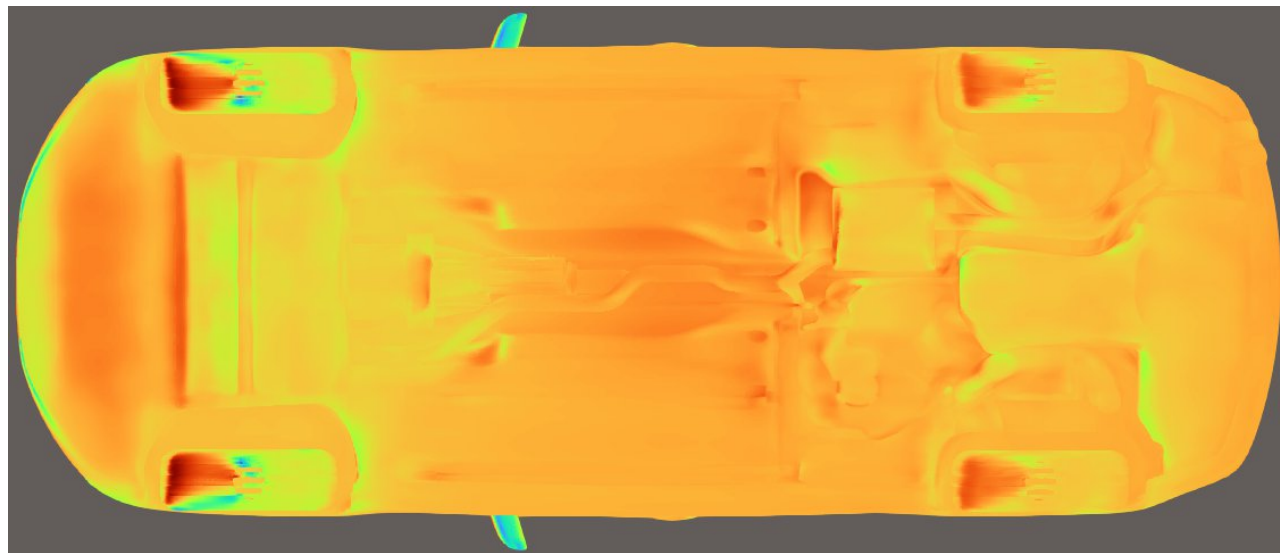
GINOT\_SATLOSS



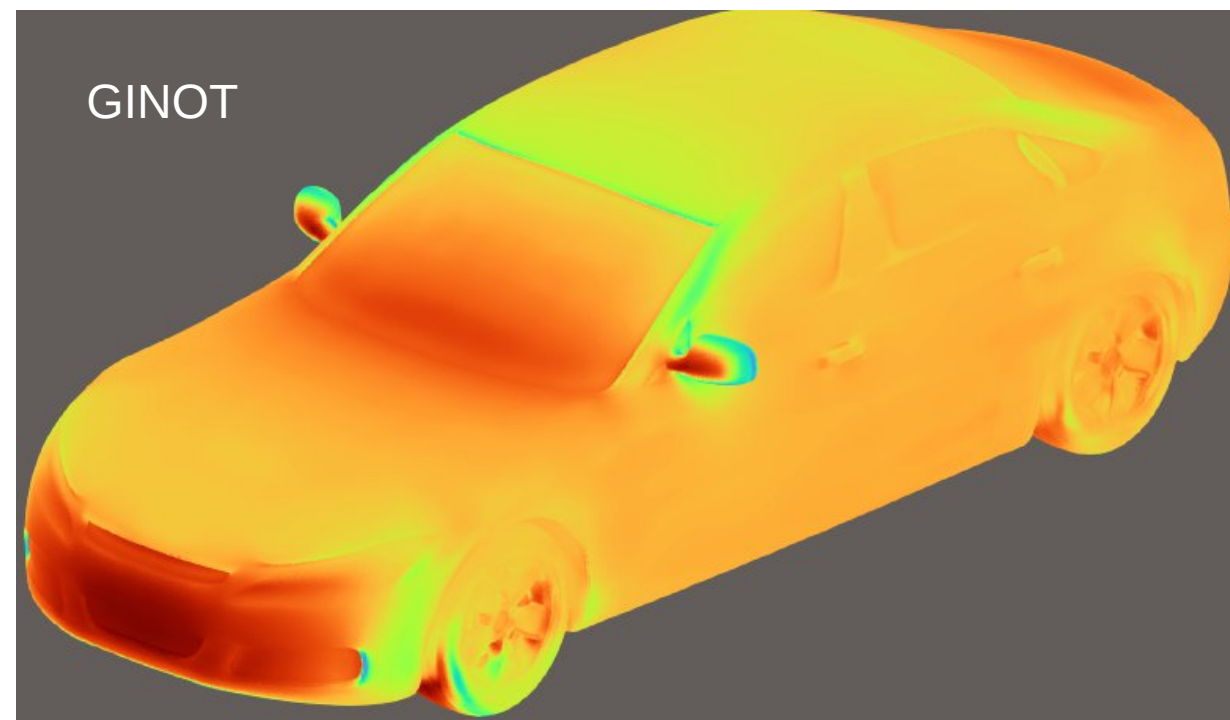
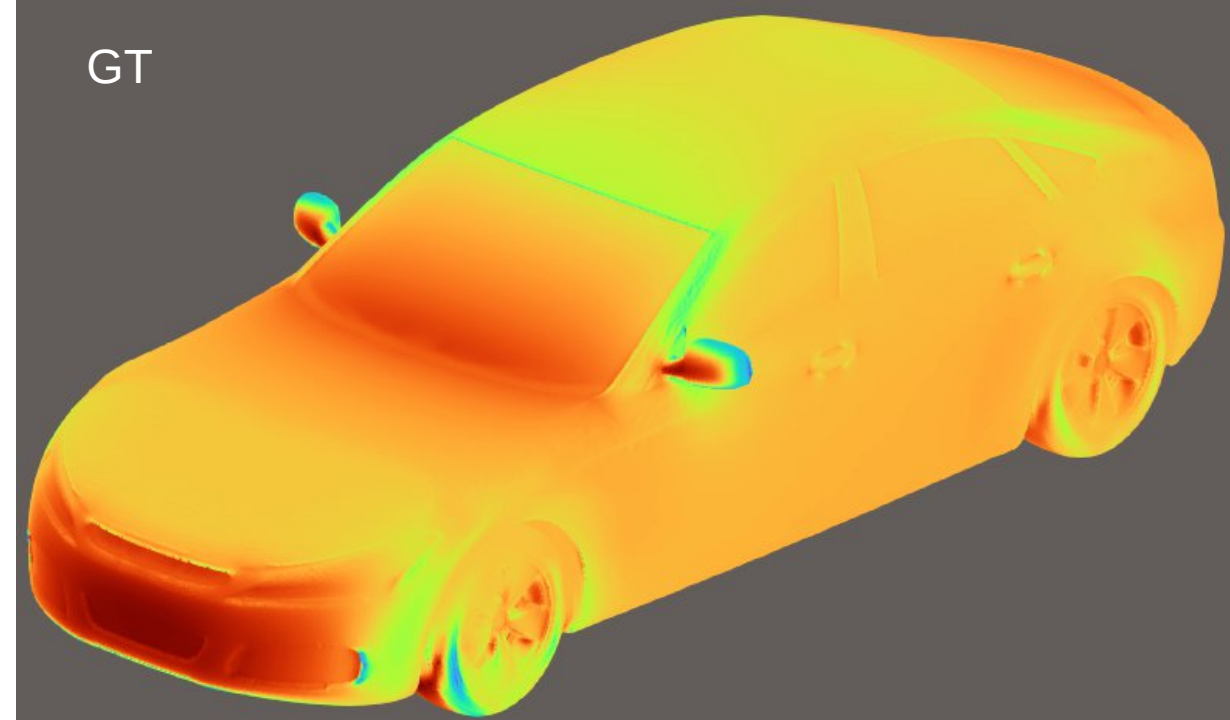
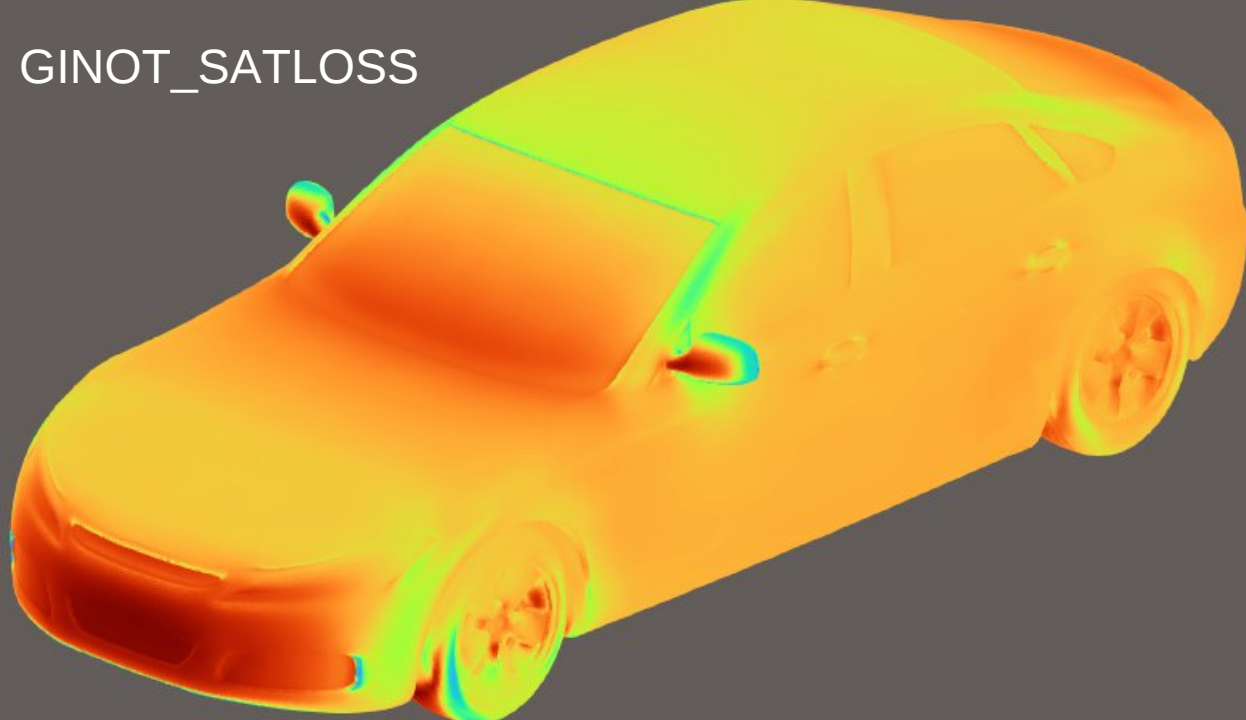
GT



GINOT







# AB-UPT

